

A Reflective Journey: Navigating My Cumulative Experience at Iowa State University

As I look back on my education at Iowa State, there are a couple things that I can look back on and be thankful for. The first of which would be my amazing peers. Throughout my entire journey here in Ames, I've always been side by side with some of the best friends I've ever had and hopefully will ever have. From studying in calculus class on a frigid Monday morning in January to playing sand volleyball in the blazing sun in August, my peers have always been right there with me every step of the way.

Another example would be that many of the courses I've taken have included some form of hands-on lab or application portion. For example, all my Cybersecurity Engineering core courses included labs dedicated to teaching me how to apply the concepts I learned in lecture. They used the Internet Scale Event and Attack Generation Environment (ISEAGE) hosted at Iowa State as a sandbox for replicating real-life networking systems for us to work with. And, like real systems, they sometimes didn't work exactly the way we expected them to. This led to many instances of going outside the scope of the class to fix or alleviate issues related to the systems so that I could complete the lab objectives.

If I had to pick a project that taught me the most about teamwork and real-life problem solving, (other than Senior Design), it would have to be the Embedded Systems course I took my sophomore year. The course was all about learning to use code, as well as a Roomba vacuum cleaner robot, to complete tasks. Writing the code wasn't too tricky, but dealing with real hardware systems was a bit of a beast. The Roombas came with a TIVA datasheet that was hundreds of pages long, which we learned how to skim and interpret for useful insights. Multiple robots had differences, such as calibration, sensor response and communication with the desktop we used to run the project. That, on top of contending with other project groups for the limited number of robots available to use each day, made it feel almost like a real design production.

Apart from my engineering courses, I was lucky enough to take two general education courses to broaden my knowledge. One of which was Russia Today, a course covering contemporary Soviet Union and Russia, from WW2 to the present. This course helped me learn about topics not covered in engineering classes, such as global politics, economics and environmental impacts. Learning about the Chernobyl nuclear meltdown was especially interesting. I also took a History of American Rock and Roll class, which highlighted musical trends, artists and bands from the turn of the 20th century until the present. This course taught me all about the societal and cultural impacts that music has had on America for over 100 years. Learning about all the different eras and effects music has had on the country was very eye-opening and made me think about the events and people that make a nation what it is today.

To go along with curricular activities, I also had the opportunity to participate in events and groups outside the classroom. One of which was the Hacking And Cybersecurity Club's (HACC) semesterly Hackathon, where teams are given misconfigured or outdated systems that they must fix and upgrade. Then, for an entire day, professional red team hackers from the industry visit and try their hand at breaking into our systems. This event helped me develop not only the technical skills needed to defend against the pros, but also soft interpersonal skills needed to document and explain our methods.

Just this last summer, I was faced with a challenge. My internship, which I had the summer prior as well as the winter in between, fell through in the last week of April. This left me with a summer with no plans, goals, or ways to improve myself. After not finding another opening by late May, I decided that I would spend my summer gaining certifications and other non-internship experience. I achieved the CompTIA Security+ certification, as well as many other smaller certs. I started my first home lab, which has proven much more time consuming and labor intensive than I once thought. And I also chose to travel to Iceland, to experience a culture and landscape that was distinctly non-Iowan.

From arriving on campus as a wide-eyed freshman who didn't know what he wanted to do the rest of his life, to a senior preparing to enter the workforce, my experience at Iowa State has been a rollercoaster of time. To my peers and friends who weathered the storm with me, I thank you for all the late nights, early mornings, and packed days in between that made my short stay in Ames a blast.